LAPORAN PENGERJAAN UJI KOMPETENSI KEJURUSAN

APLIKASI SMART PARKING

Disusun untuk Memenuhi Sebagai Persyaratan Kelulusan Uji Kompetensi Keahlian

Tahun Pelajaran 2025/2026

DISUSUN OLEH :

Cellomitha Halawah NIS. 3224.10.354

PROGRAM KEAHLIAN PENGEMBANGAN PERANGKAT LUNAK DAN GIM KOMPETENSI KEAHLIAN REKAYASA PERANGKAT LUNAK

SMKS MAHARDHIKA BATUJAJAR

2025/2026

LAPORAN PENGERJAAN UJI KOMPETENSI KEJURUAN

SMK MAHARDHIKA BATUJAJAR

TAHUN PELAJARAN 2025-2026

KOPETENSI KEAHLIAN : Rekayasa Perangkat Lunak

KELAS : XII RPL 1

NIS : 3224.10.354

PAKET SOAL : Paket 1 (Smart Parking)

JENIS APLIKASI : Website

ALOKASI WAKTU : 12 Jam

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a. PENJADWALAN KEGIATAN

Kegiatan	Alokasi Waktu																			
	Minggu 1			Minggu 2			Minggu 3			Minggu 4			Minggu 5				Minggu 6			
	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Analisis Soal	■	■																		
Perencanaan Sistem			■	■																
Perancangan DB					■															
Perancangan UI/UX						■	■													
Implementasi DB								■												
Implementasi UI/UX									■	■										
Implementasi Program											■	■	■	■	■					
Finalisasi Program																■				
Pengujian Program																	■			
Hosting + Github																		■		
Finalisasi Dokumentasi																			■	
Finalisasi Projek UKK																				■
Dokumentasi	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■

b. PERSIAPAN PERANGKAT KERAS

No	Hardware	Fungsi
1.	Laptop	Digunakan untuk membuat, menjalankan, dan mengelola aplikasi parkir serta database.
2.	ESP32	Digunakan sebagai mikrokontroler utama untuk menghubungkan perangkat RFID dengan sistem parkir dan mengirim data ke server.
3.	Kabel USB Type C	Digunakan untuk menghubungkan ESP32 dengan komputer saat proses upload program.
4.	RFID Module Reader RC522 + Tag	Digunakan untuk membaca kartu RFID yang digunakan sebagai identitas kendaraan saat melakukan check-in dan check-out parkir.
5.	Servo Motor SG90	Digunakan sebagai penggerak palang parkir yang akan terbuka dan tertutup secara otomatis.
6.	LCD 16x2 I2C	Digunakan untuk menampilkan informasi seperti status parkir, pesan akses diterima, atau akses ditolak.
7.	Kabel Jumper	Digunakan untuk menghubungkan komponen elektronik seperti ESP32, RFID, LCD, dan servo motor

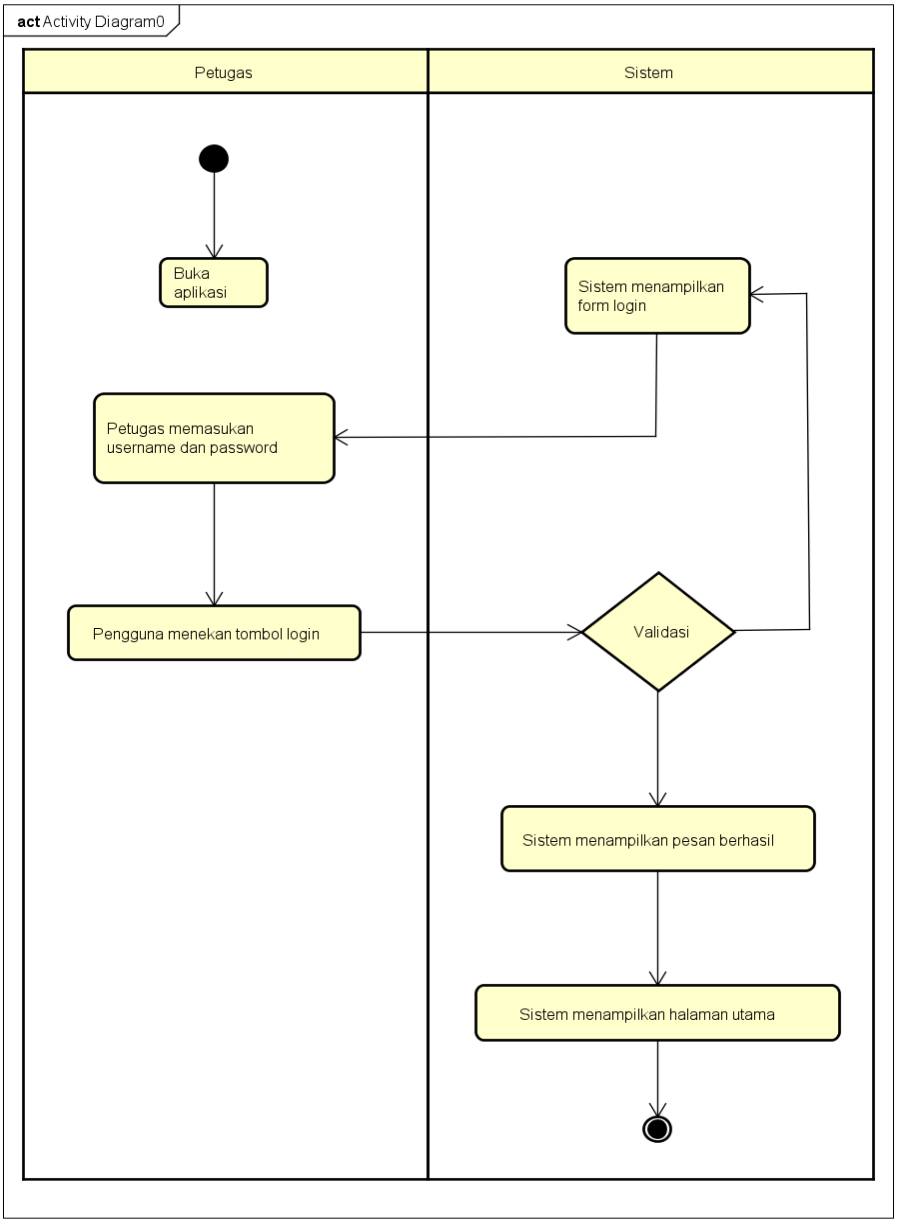
c. PERSIAPAN PERANGKAT LUNAK

No	**Software**	**Fungsi**
1.	PC / Laptop	Digunakan sebagai perangkat utama untuk membuat program, menjalankan aplikasi parkir, serta menghubungkan perangkat keras dengan sistem.
2.	Koneksi Internet	Digunakan untuk menghubungkan perangkat IoT dengan server melalui jaringan internet sehingga data parkir dapat dikirim dan diterima secara realtime.
3.	Sistem Operasi – Microsoft Windows	Digunakan sebagai sistem operasi pada komputer yang mendukung proses pengembangan dan menjalankan aplikasi parkir.
4.	Text Editor / Tools Pemrograman – Visual Studio Code	Digunakan untuk menulis dan mengedit kode program seperti HTML, CSS, JavaScript, dan PHP pada aplikasi parkir.
5.	Aplikasi Server Basis Data – XAMPP	Digunakan sebagai server lokal untuk menjalankan aplikasi web serta mengelola database sistem parkir menggunakan MySQL.
6.	Aplikasi MQTT Client – HiveMQ WebSocket Client	Digunakan untuk menguji komunikasi MQTT antara perangkat IoT dan sistem parkir.
7.	Debugging Tools – Arduino IDE	Digunakan untuk menulis, mengupload, dan menguji program pada mikrokontroler ESP32.

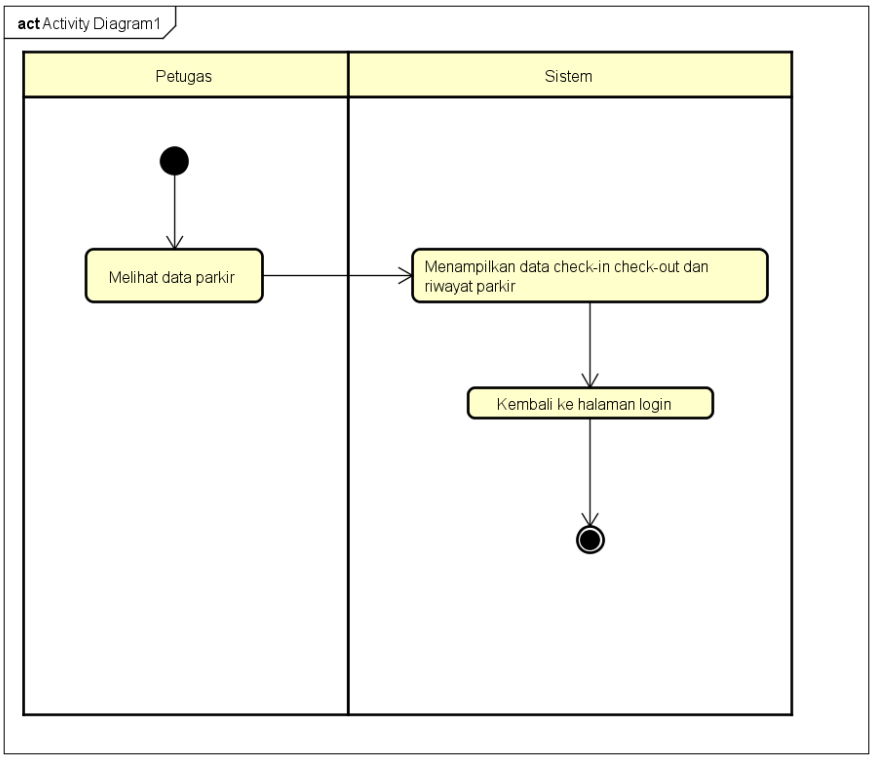
8.	Aplikasi Pengolahan Data – Microsoft Word	Digunakan untuk membuat dan menyusun laporan proyek aplikasi parkir.

d. PERANCANGAN ALGORITMA APLIKASI

Activity Diagram login



Activity Diagram Melihat



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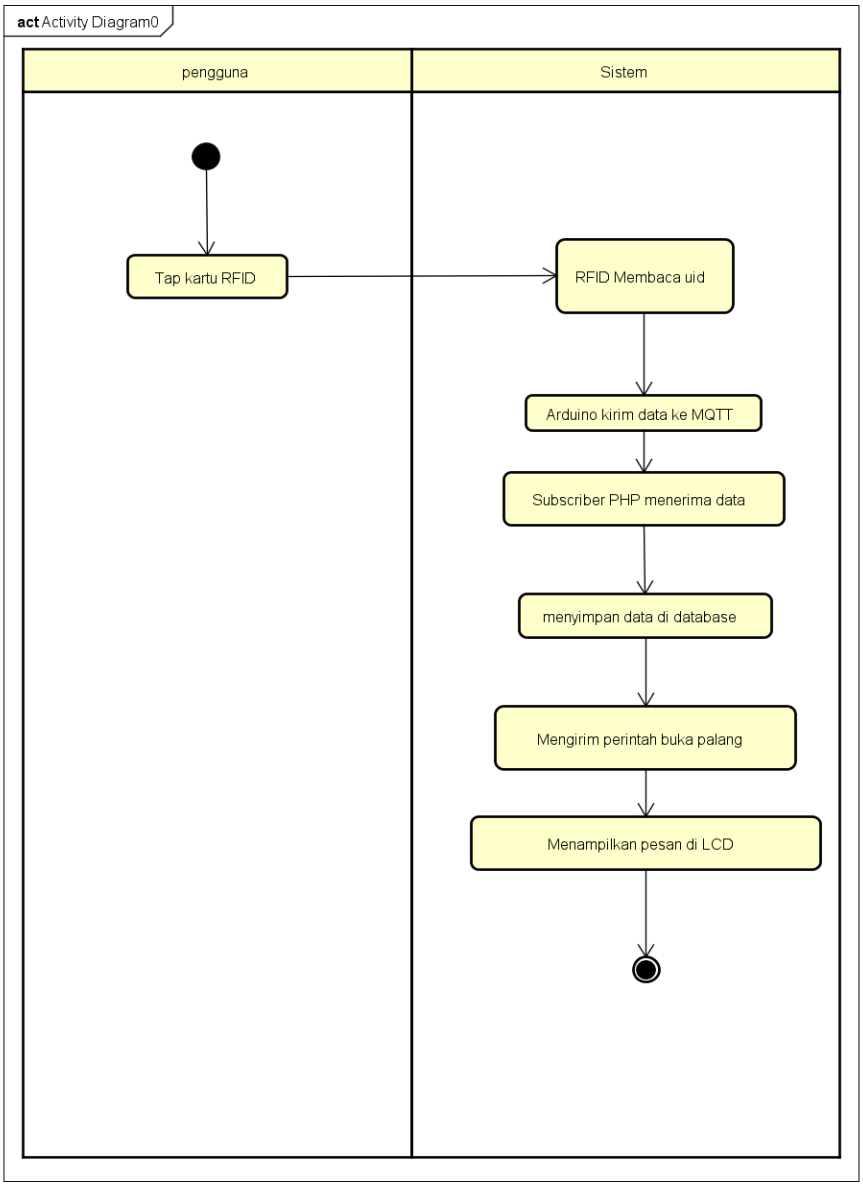
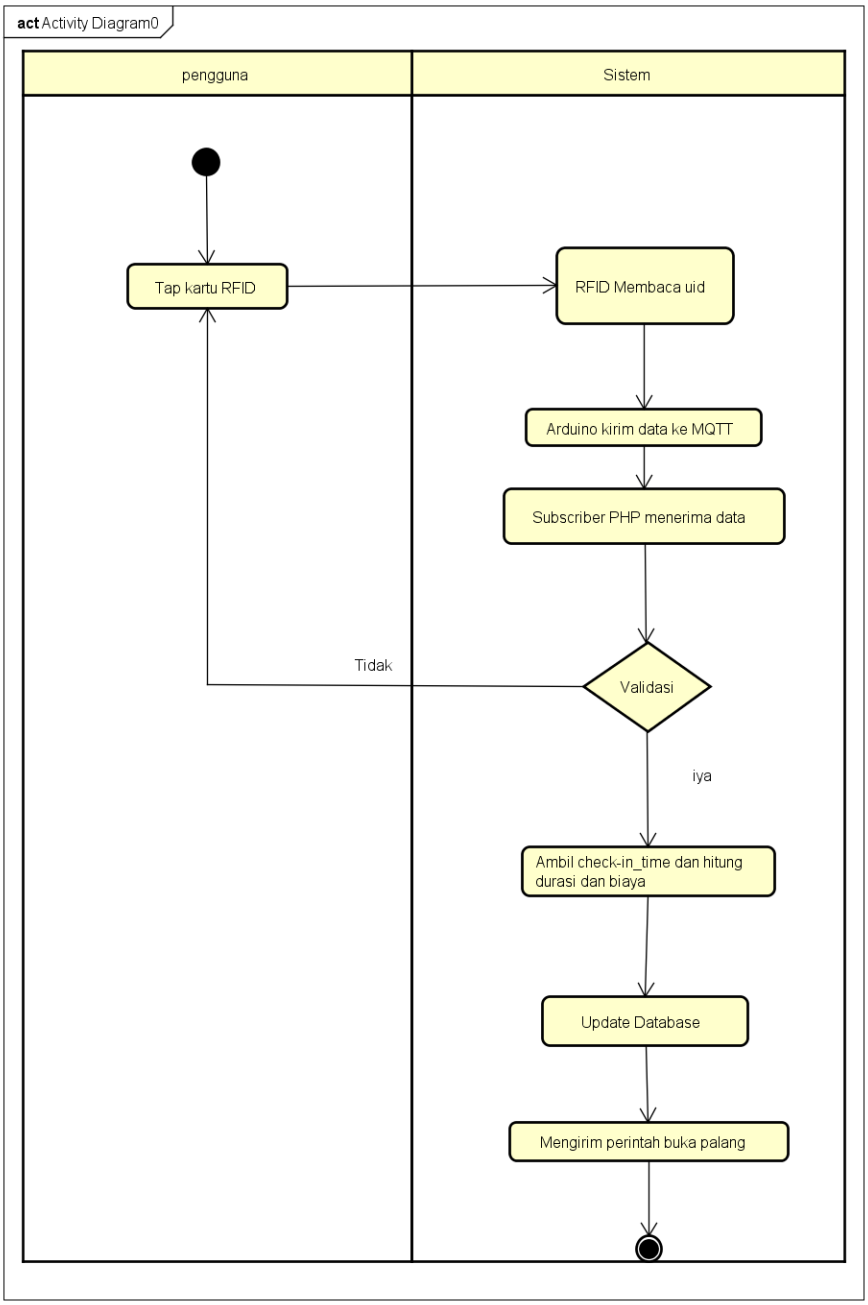
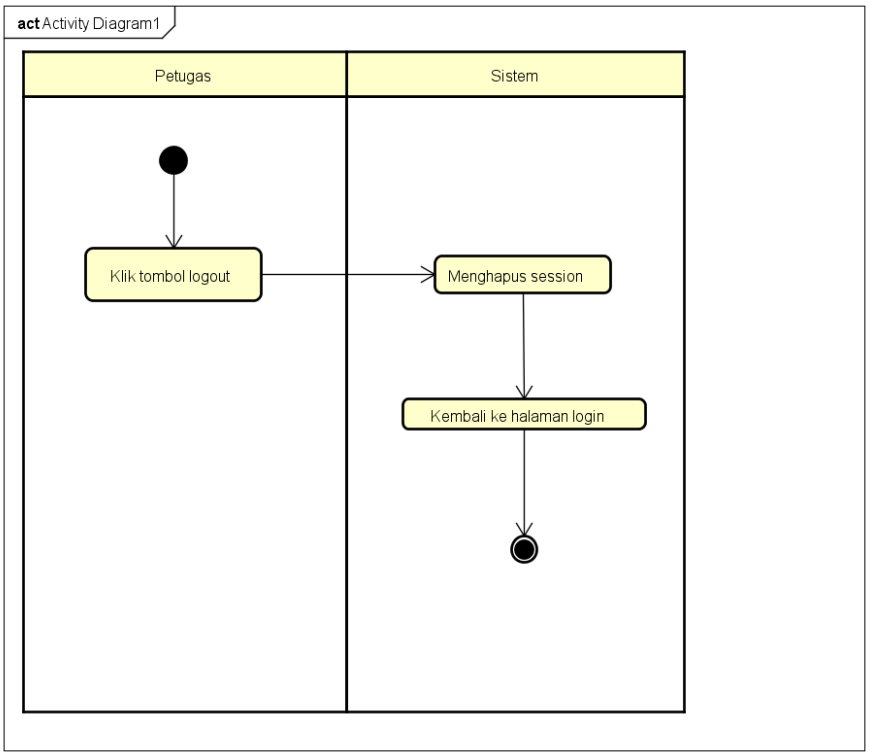
Activity Diagram Checkin



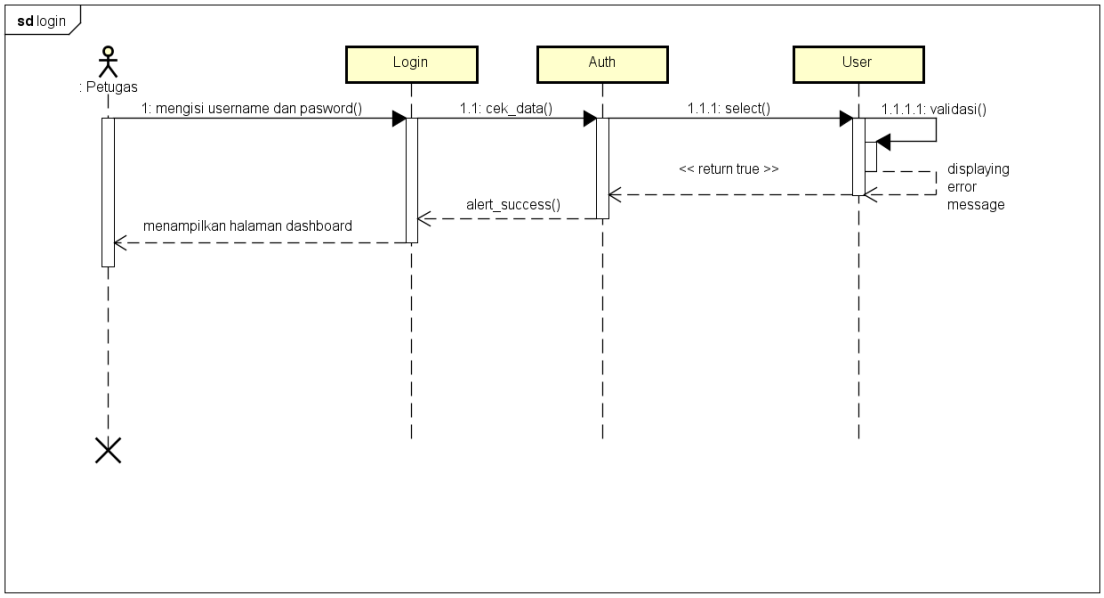
Activity Diagram Checkout

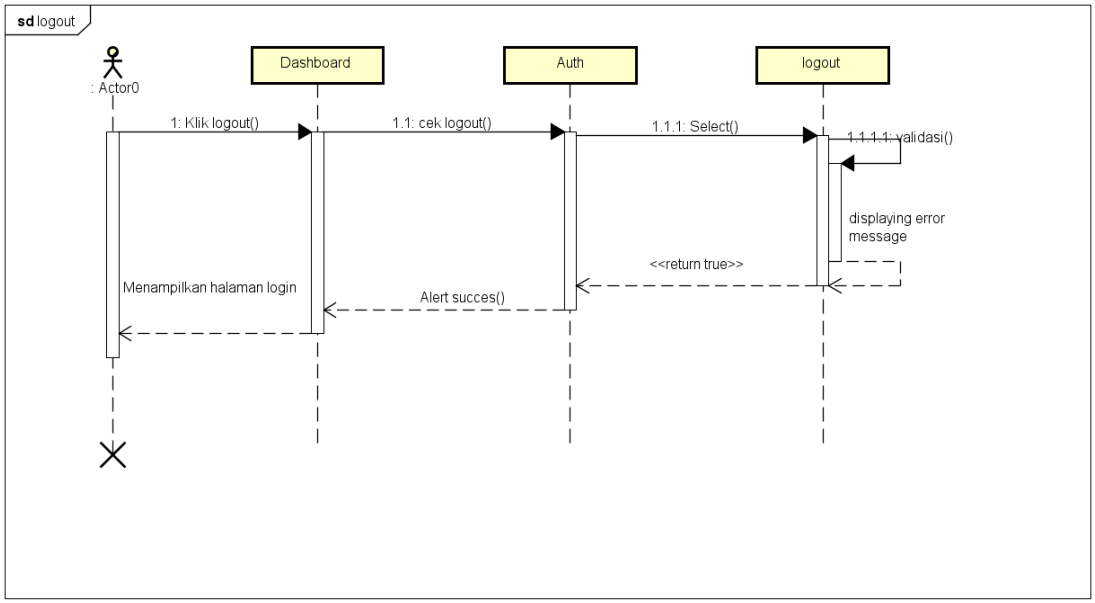


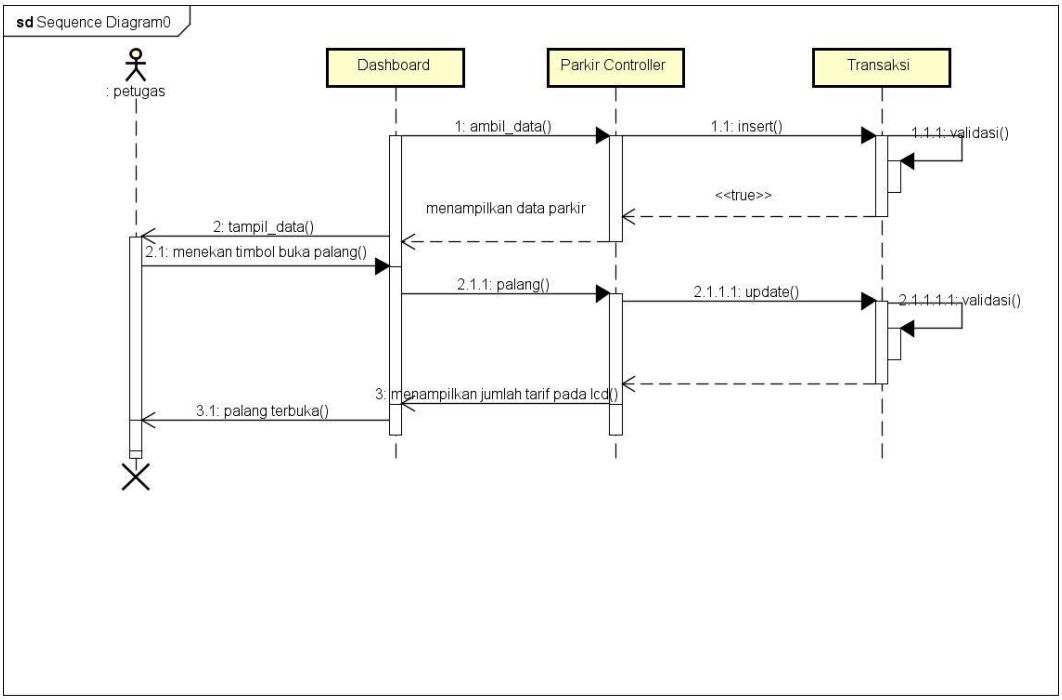
Activity Diagram Logout



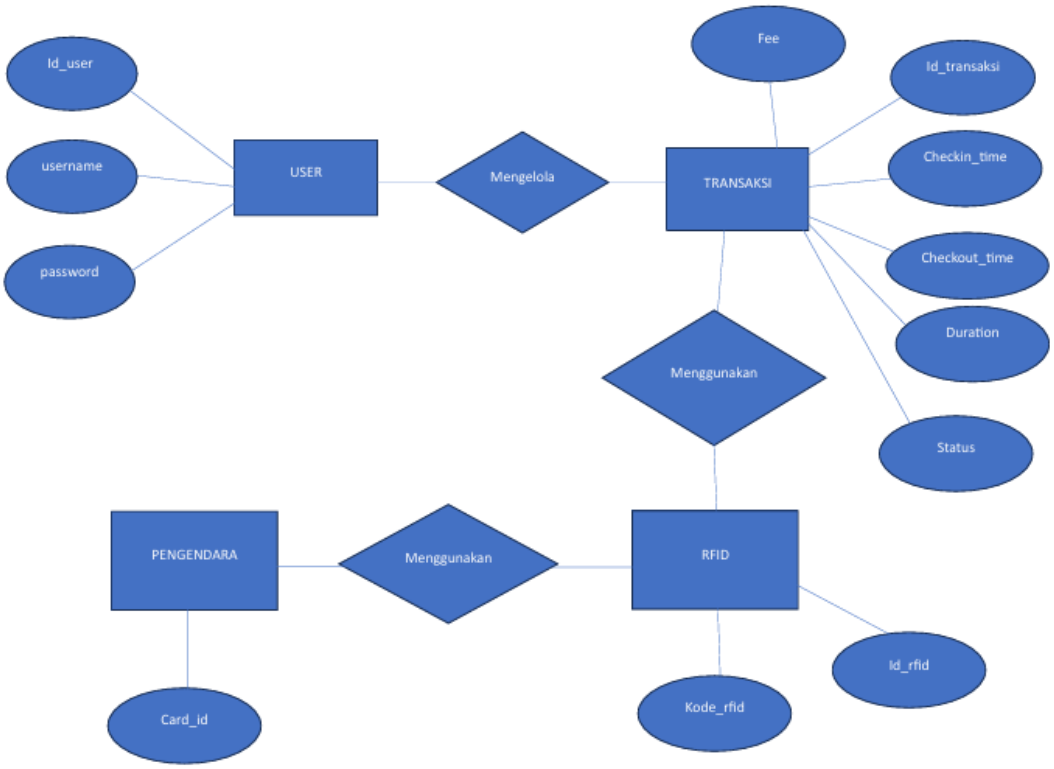
e. SEQUENCE DIAGRAM





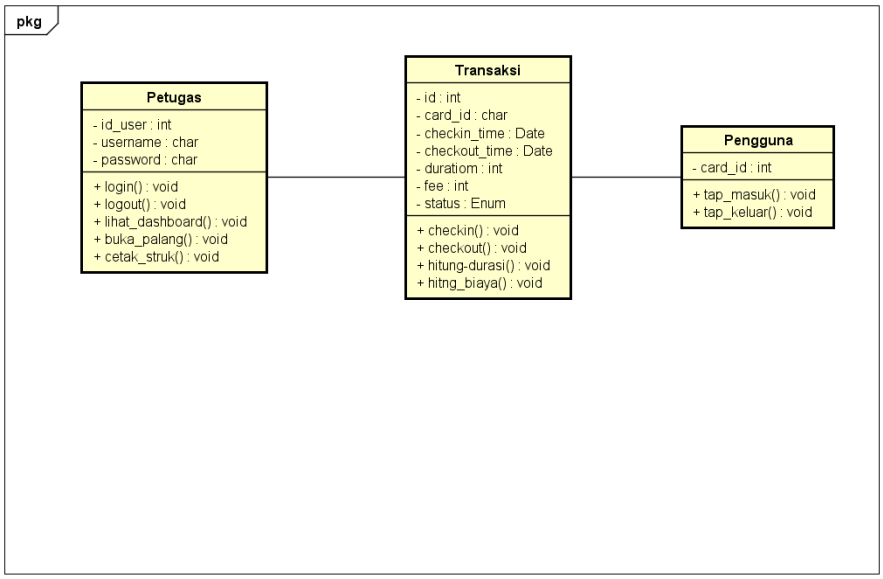

sd Sequence Diagram0
: petugas
Dashboard
Parkir Controller
Transaksi
1: ambil_data()
1.1: insert()
1.1.1: validasi()
<<true>>
menampilkan data parkir
2: tampil_data()
2.1: menekan timbol buka palang()
2.1.1: palang()
2.1.1.1: update()
2.1.1.1.1: validasi()
3: menampilkan jumlah tarif pada lcd()
3.1: palang terbuka()

f. ERD

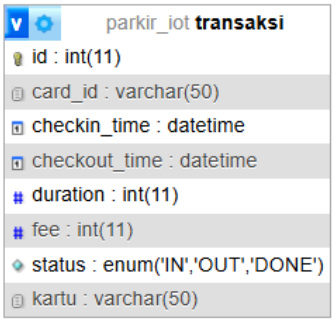

Id_user
username
password
USER
Mengelola
TRANSAKSI
Fee
Id_transaksi
Checkin_time
Checkout_time
Duration
Status
Menggunakan
PENGENDARA
Menggunakan
RFID
Card_id
Kode_rfid
Id_rfid

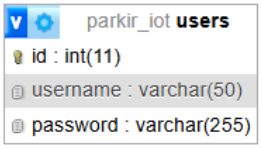
g. PERANCANGAN BASIS DATA

CLASS DIAGRAM



DESAIN





h. DUMMY DATA

Data Tabel User

			id	username	password
Edit	Copy	Delete	1	petugas	827ccb0eea8a706c4c34a16891f84e7b
Edit	Copy	Delete	2	cello	827ccb0eea8a706c4c34a16891f84e7b

Data Tabel Transaksi

			id	card_id	checkin_time	checkout_time	duration	fee	status	kartu
Edit	Copy	Delete	13	1234	2026-03-18 18:04:17	2026-03-18 18:04:30	1	2000	OUT	NULL
Edit	Copy	Delete	14	11223344	2026-03-25 23:04:01	2026-03-25 23:06:30	3	2000	OUT	NULL
Edit	Copy	Delete	15	1234	2026-03-25 23:05:35	2026-03-25 23:05:43	1	2000	OUT	NULL

i. PEMBUATAN BASIS DATA (MySQL Code)

Tabel Users

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
1	**id**	int(11)			No	*None*		AUTO_INCREMENT	Change Drop More
2	**username**	varchar(50)	latin1_swedish_ci		Yes	*NULL*			Change Drop More
3	**password**	varchar(255)	latin1_swedish_ci		Yes	*NULL*			Change Drop More

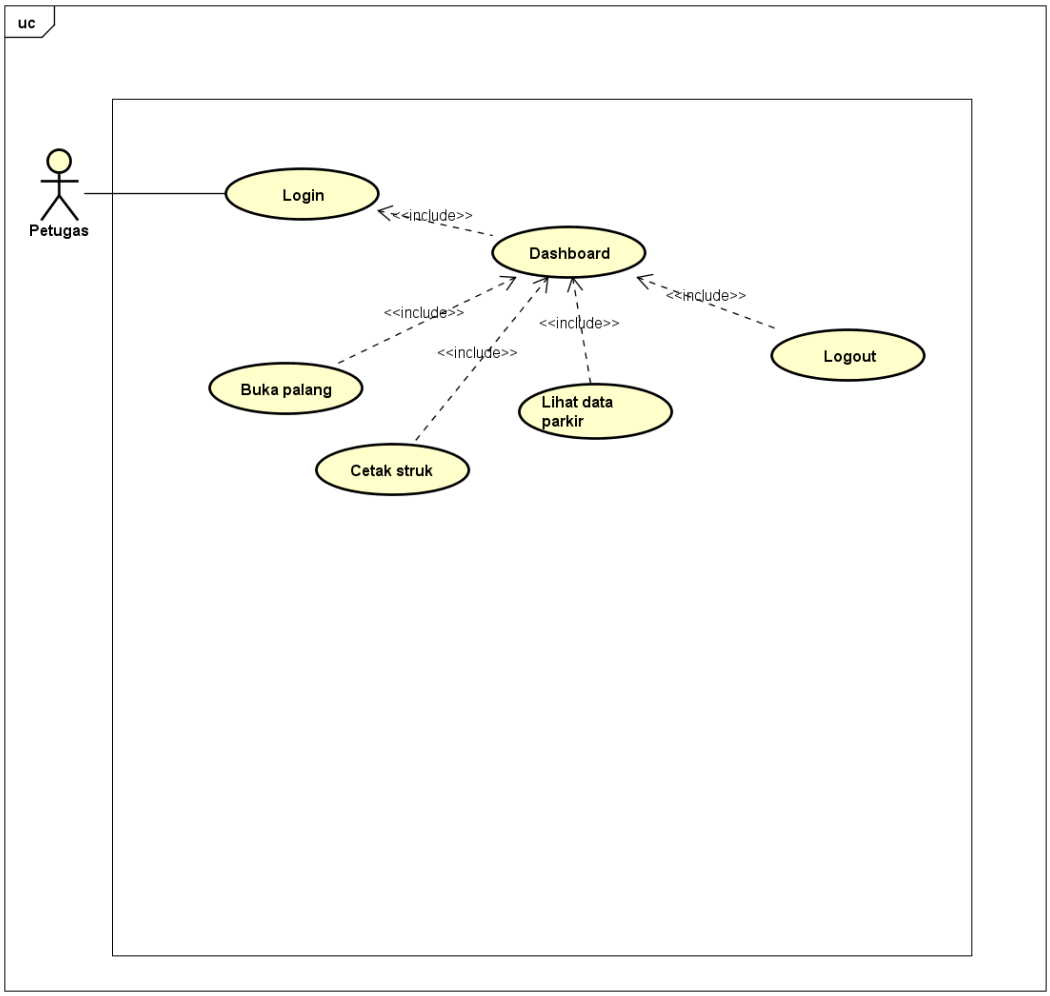
Tabel Transaksi

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
1	**id**	int(11)			No	*None*		AUTO_INCREMENT	Change Drop More
2	**card_id**	varchar(50)	latin1_swedish_ci		Yes	*NULL*			Change Drop More
3	**checkin_time**	datetime			Yes	*NULL*			Change Drop More
4	**checkout_time**	datetime			Yes	*NULL*			Change Drop More
5	**duration**	int(11)			Yes	*NULL*			Change Drop More
6	**fee**	int(11)			Yes	*NULL*			Change Drop More
7	**status**	enum('IN', 'OUT', 'DONE')	latin1_swedish_ci		Yes	*NULL*			Change Drop More
8	**kartu**	varchar(50)	latin1_swedish_ci		Yes	*NULL*			Change Drop More

j. PERANCANGAN TAMPILAN ANTARMUKA (UI) WEB

Aplikasi Smart Parking

a. Use Case Diagram



b. Skenario Use Case

Name : Login

IDE : UC001

Description : Memasukan Akun

Actors : Petugas

Main Cours	
Aksi Aktor	**Reaksi Sistem**
1. Membuka halaman login	
	2. Menampilkan halaman login
3. Memasukan username dan password	
	4. Memvalidasi data logim
	5. Menampilkan halaman dashboard

Alternate Cours	
Aksi Aktor	**Reaksi Sistem**
1. Membuka halaman login	
	2. Menampilkan halaman login
3. Memasukan username dan password yang tidak valid	
	4. Menampilkan pesan error

Name : Logout

IDE : UC002

Description : Keluar Akun

Actors : Petugas

Main Cours	
Aksi Aktor	**Reaksi Sistem**
1. Berada pada halaman dashboard	
2. Menekan tombol logout	
	3. Memperoses permintaan logout
	4. Mengarahkan Kembali ke halaman login

Alternate Cours	
Aksi Aktor	**Reaksi Sistem**
1. Terjadi kesalahan	
	2. Menampilkan pesan error
3. Coba kembali	

Name : Check-in

IDE : UC001

Description : Masuk parkir

Actors : Pengendara

Main Cours	
Aksi Aktor	**Reaksi Sistem**
1. Pengendara memasuki area parkir	
2. Tempelkan kartu atau id kendaraan	
	3. Sensor/RFID membaca ID kendaraan
	4. Menerima data kendaraan
	5. Menyimpan data kendaraan dan jam masuk ke database
	6. Menampilkan kendaraan pada dashboard parkir

Alternate Cours	
Aksi Aktor	**Reaksi Sistem**
1. Tempelkan kartu atau id kendaraan	
	2. Gagal membaca kartu
	3. Menampilkan pesan error
4. Tap ulang kartu	

Name : Check-out

IDE : UC001

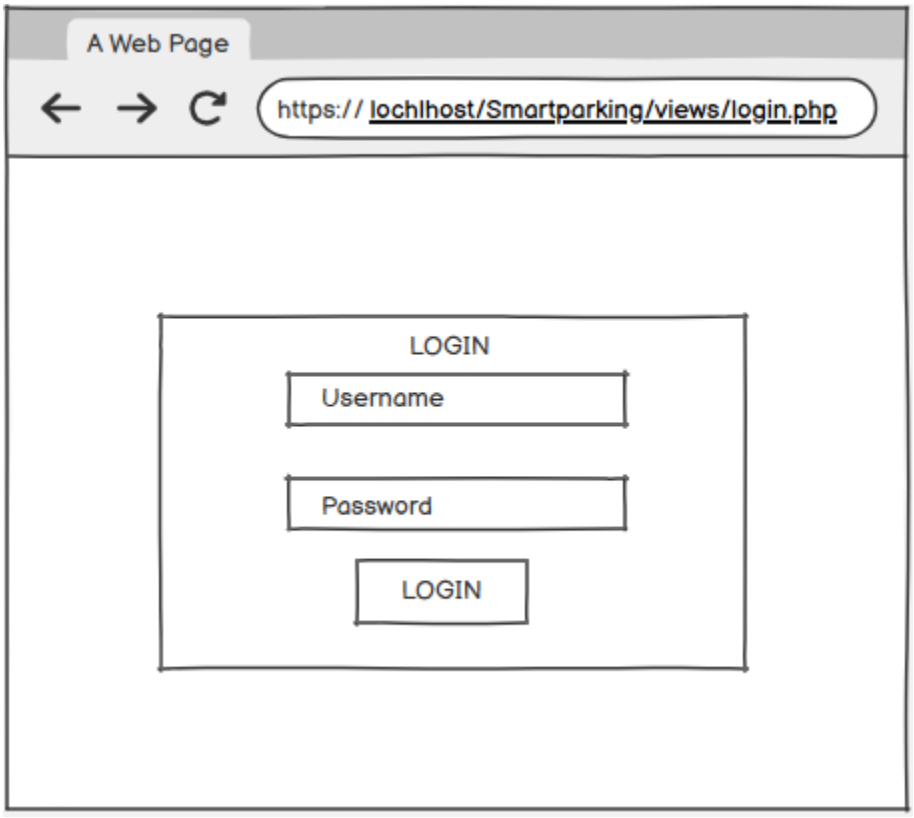
Description : Keluar parkir

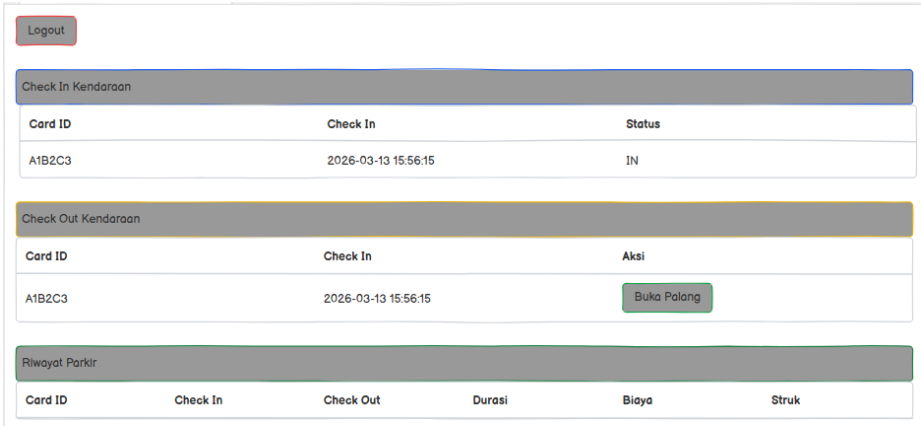
Actors : Pengendara

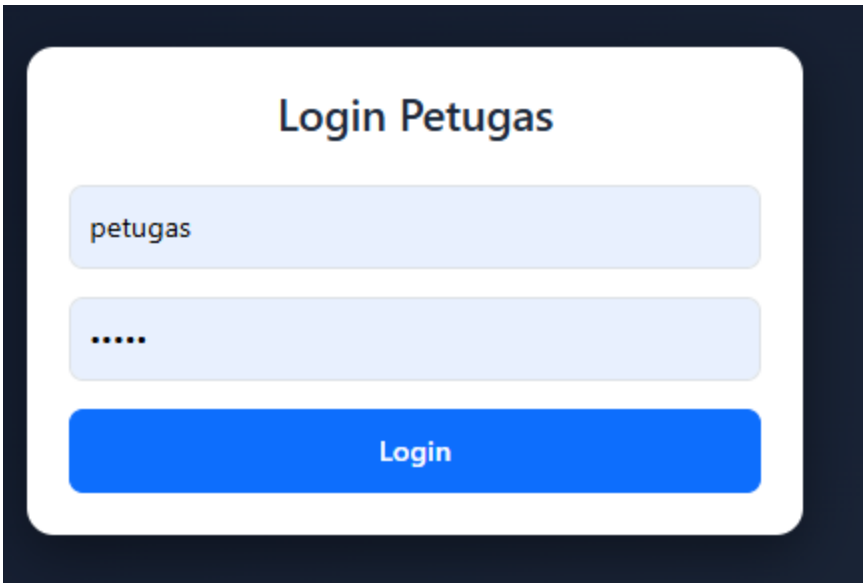
Main Cours	
Aksi Aktor	**Reaksi Sistem**
1. Tap kartu untuk keluar	
	2. Sensor / RFID membaca ID kendaraan
	3. Menerima data kendaraan
	4. Mengambil data jam masuk dari database
	5. Menghitung durasi dan biaya parkir

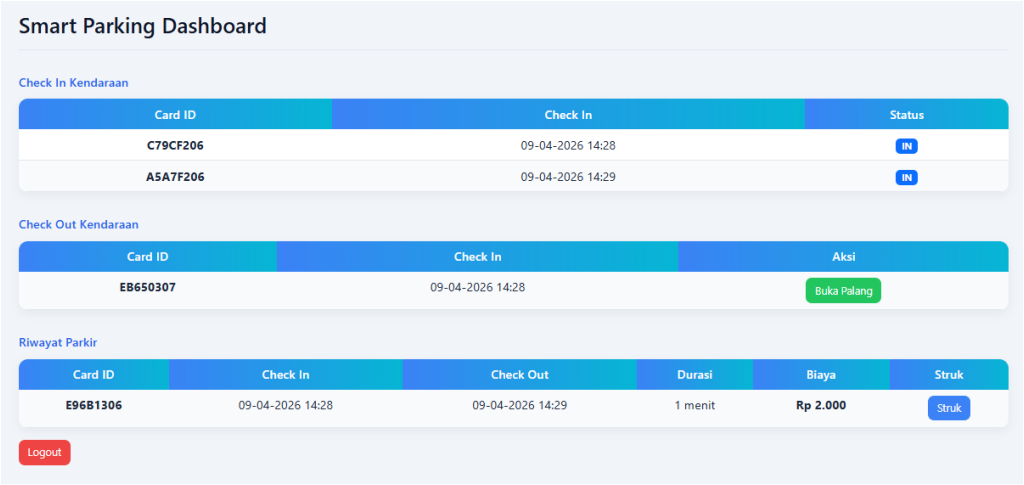
Alternate Cours	
Aksi Aktor	**Reaksi Sistem**
1. Tap kartu untuk keluar	
	2. Id tidak ditemukan
	3. Menampilkan pesan error
4. Tap ulang kartu	

C. Mock Up Aplikasi









Smart Parking Dashboard

Check In Kendaraan

Card ID	Check In	Status
C79CF206	09-04-2026 14:28	IN
A5A7F206	09-04-2026 14:29	IN

Check Out Kendaraan

Card ID	Check In	Aksi
EB650307	09-04-2026 14:28	Buka Palang

Riwayat Parkir

Card ID	Check In	Check Out	Durasi	Biaya	Struk
E96B1306	09-04-2026 14:28	09-04-2026 14:29	1 menit	Rp 2.000	Struk

Logout

k. PEMBUATAN STRUKTUR WEB (PHP)

a. File Login. Php

```
<!DOCTYPE html>
<html>
<head>

<title>Login Petugas</title>

<link
href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.2/dist/css/bootstr
ap.min.css" rel="stylesheet">

<style>
body {
    background: linear-gradient(135deg, #0f172a, #1e293b);
    height: 100vh;
}

/* Card */
.card {
    border: none;
    border-radius: 15px;
    box-shadow: 0 10px 30px rgba(0,0,0,0.4);
}

/* Input */
.form-control {
    border-radius: 8px;
    padding: 10px;
}

/* Button */
.btn-primary {
    border-radius: 8px;
    padding: 10px;
    font-weight: 500;
}

/* Judul */
h4 {
    font-weight: 600;
    color: #1e293b;
}

/* Center full */
.full-height {
    height: 100vh;
}
</style>

</head>

<body>
```

```
<div class="container full-height d-flex align-items-center
justify-content-center">

<div class="col-md-4">

<div class="card p-4">

<h4 class="text-center mb-4"> Login Petugas</h4>

<form action="../controllers/AuthController.php" method="POST">

<input name="username" class="form-control mb-3"
placeholder="Username" required>

<input type="password" name="password" class="form-control mb-3"
placeholder="Password" required>

<button class="btn btn-primary w-100">Login</button>

</form>

</div>

</div>

</div>

</body>
</html>
```

b. Buka_palang. Php

```
<?php
include "config/database.php";

// ambil id dari URL
$id = $_GET['id'] ?? 0;

// ambil data transaksi
$q = $conn->query("SELECT * FROM transaksi WHERE id='$id'");
$row = $q->fetch_assoc();

// validasi data
if(!$row){
    die("Data tidak ditemukan");
}

// waktu masuk & keluar
$masuk = strtotime($row['checkin_time']);
$keluar = time();
```

```
        // hitung selisih (detik)
        $selisih = $keluar - $masuk;

        // biar ga minus
        if($selisih < 0){
            $selisih = 0;
        }

        // durasi menit
        $durasi = ceil($selisih / 60);

        // MINIMAL 1 MENIT BIAR GA 0
        if($durasi <= 0){
            $durasi = 1;
        }

        // tarif per jam
        $tarif = 2000;

        // hitung biaya (minimal 1 jam)
        $jam = ceil($durasi / 60);
        if($jam <= 0){
            $jam = 1;
        }

        $biaya = $jam * $tarif;


        // ================= UPDATE DATABASE =================
        $conn->query("UPDATE transaksi SET
            checkout_time = NOW(),
            duration = '$durasi',
            fee = '$biaya',
            status = 'OUT'
        WHERE id='$id'");


        // ================= MQTT =================
        require_once("mqtt/phpMQTT.php");
        use Bluerhinos\phpMQTT;

        $server = "broker.hivemq.com"; // HARUS SAMA DENGAN ESP32
        $port = 1883;
        $client_id = "server-" . rand();

        $mqtt = new phpMQTT($server, $port, $client_id);

        if($mqtt->connect()){

            //  PESAN LCD
            $pesan = "Rp " . number_format($biaya,0,',','.');

            // tampilkan ke LCD
            $mqtt->publish("parking/KELOMPOK0122/lcd", $pesan, 0);

            // buka palang
```

```
            $mqtt->publish("parking/KELOMPOK0122/exit/servo", "OPEN",
        0);

            $mqtt->close();
        }


        // ================= REDIRECT =================
        header("Location: views/dashboard.php");
exit;
```

c. Dashboard. php

```
        <?php
        include "../controllers/DashboardController.php";
        ?>

        <link
        href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.2/dist/css/b
        ootstrap.min.css" rel="stylesheet">

        <div class="container mt-4">

        <h3>Dashboard Smart Parking</h3>

        <a href="../views/login.php" class="btn btn-danger mb-
        4">Logout</a>


        <!-- CHECK IN -->
        <div class="card mb-4">
        <div class="card-header bg-primary text-white">
        Check In Kendaraan
        </div>

        <div class="card-body">

        <table class="table table-bordered">
        <thead>
        <tr>
        <th>Card ID</th>
        <th>Check In</th>
        <th>Status</th>
        </tr>
        </thead>

        <tbody>

        <?php while($row = $data->fetch_assoc(){ ?>
```

```
<tr>

<td><?= $row['card_id'] ?></td>
<td><?= $row['checkin_time'] ?></td>
<td><?= $row['status'] ?></td>

</tr>

<?php } ?>

</tbody>
</table>

</div>
</div>



<!-- CHECK OUT -->
<div class="card mb-4">
<div class="card-header bg-warning">
Check Out Kendaraan
</div>

<div class="card-body">

<table class="table table-bordered">

<tr>
<th>Card ID</th>
<th>Check In</th>
<th>Aksi</th>
</tr>

<?php while($k=$keluar->fetch_assoc()){ ?>

<tr>
<td><?= $k['card_id']?></td>
<td><?= $k['checkin_time']?></td>

<td>
<a href="../buka_palang.php?id=<?=$k['id']?>" class="btn btn-
success btn-sm">
Buka Palang
</a>
</td>

</tr>

<?php } ?>

</table>

</div>
</div>
```

```
<!-- RIWAYAT -->
<div class="card">

<div class="card-header bg-success text-white">
Riwayat Parkir
</div>

<div class="card-body">

<table class="table table-bordered">

<tr>
<th>Card ID</th>
<th>Check In</th>
<th>Check Out</th>
<th>Durasi</th>
<th>Biaya</th>
<th>Struk</th>
</tr>

<?php while($r=$riwayat->fetch_assoc()){ ?>

<tr>

<td><?= $r['card_id']?></td>
<td><?= $r['checkin_time']?></td>
<td><?= $r['checkout_time']?></td>
<td><?= $r['duration']?></td>
<td><?= $r['fee']?></td>

<td>
<a href="../struk.php?id=<?=$r['id']?>" class="btn btn-
success btn-sm">
Struk
</a>
</td>

</tr>

<?php } ?>

</table>

</div>
</div>


</div>
```

d. Subscriber. Php

```
<?php
date_default_timezone_set('Asia/Jakarta');

require_once("phpMQTT.php");
use Bluerhinos\phpMQTT;

include "../config/database.php";

$server = "broker.hivemq.com";
$port = 1883;
$client_id = "phpMQTT-subscriber";

$mqtt = new phpMQTT($server,$port,$client_id);

if(!$mqtt->connect()){
    exit(1);
}

echo "Connected to broker\n";

// FIX TOPIC
$topics['parking/KELOMPOK0122/entry/rfid'] =
array("qos"=>0,"function"=>"procmsg");
$topics['parking/KELOMPOK0122/exit/rfid'] =
array("qos"=>0,"function"=>"procmsg");

$mqtt->subscribe($topics,0);

while($mqtt->proc()){
}

$mqtt->close();


function procmsg($topic, $msg){

global $conn;

$data = json_decode($msg, true);
$card = $data['rfid'] ?? '';

if($card == '') return;


// ================= ENTRY =================
if(strpos($topic, 'entry') !== false){

    $conn->query("INSERT INTO
transaksi(card_id,checkin_time,status)
    VALUES('$card',NOW(),'IN')");

    echo "CHECKIN\n";
}
```

```
        // ================= EXIT =================
        elseif(strpos($topic, 'exit') !== false){

            $q = $conn->query("SELECT * FROM transaksi
            WHERE card_id='$card' AND status='IN' LIMIT 1");

            if($q->num_rows > 0){

                echo "MENUNGGU PETUGAS\n";

            } else {
                echo "DATA TIDAK ADA\n";
            }
        }
}
```

e. phpMQTT. Php

```
        <?php

        namespace Bluerhinos;

        /*
              phpMQTT
              A simple php class to connect/publish/subscribe to an
        MQTT broker

        */

        /*
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*/

/* phpMQTT */

class phpMQTT
{
    protected $socket;            /* holds the socket */
    protected $msgid = 1;            /* counter for message
id */
    public $keepalive = 10;        /* default keepalive
timmer */
    public $timesinceping;        /* host unix time, used to
detect disconects */
    public $topics = [];    /* used to store currently
subscribed topics */
    public $debug = false;        /* should output debug
messages */
    public $address;            /* broker address */
    public $port;                /* broker port */
    public $clientid;            /* client id sent to brocker
*/
    public $will;                /* stores the will of the
client */
    protected $username;            /* stores username */
    protected $password;            /* stores password */

    public $cafile;
    protected static $known_commands = [
        1 => 'CONNECT',
        2 => 'CONNACK',
        3 => 'PUBLISH',
        4 => 'PUBACK',
        5 => 'PUBREC',
        6 => 'PUBREL',
        7 => 'PUBCOMP',
        8 => 'SUBSCRIBE',
        9 => 'SUBACK',
        10 => 'UNSUBSCRIBE',
        11 => 'UNSUBACK',
        12 => 'PINGREQ',
        13 => 'PINGRESP',
```

```
            14 => 'DISCONNECT'
        ];

        /**
         * phpMQTT constructor.
         *
         * @param $address
         * @param $port
         * @param $clientid
         * @param null $cafile
         */
        public function __construct($address, $port, $clientid,
    $cafile = null)
        {
            $this->broker($address, $port, $clientid, $cafile);
        }

        /**
         * Sets the broker details
         *
         * @param $address
         * @param $port
         * @param $clientid
         * @param null $cafile
         */
        public function broker($address, $port, $clientid,
    $cafile = null): void
        {
            $this->address = $address;
            $this->port = $port;
            $this->clientid = $clientid;
            $this->cafile = $cafile;
        }

        /**
         * Will try and connect, if fails it will sleep 10s and
    try again, this will enable the script to recover from a
    network outage
         *
         * @param bool $clean - should the client send a clean
    session flag
         * @param null $will
         * @param null $username
         * @param null $password
         *
         * @return bool
         */
        public function connect_auto($clean = true, $will = null,
    $username = null, $password = null): bool
        {
            while ($this->connect($clean, $will, $username,
    $password) === false) {
                sleep(10);
            }
            return true;
        }
```

```
    /**
     * @param bool $clean - should the client send a clean
session flag
     * @param null $will
     * @param null $username
     * @param null $password
     *
     * @return bool
     */
    public function connect($clean = true, $will = null,
$username = null, $password = null): bool
    {
        if ($will) {
            $this->will = $will;
        }
        if ($username) {
            $this->username = $username;
        }
        if ($password) {
            $this->password = $password;
        }

        if ($this->cafile) {
            $socketContext = stream_context_create(
                [
                    'ssl' => [
                        'verify_peer_name' => true,
                        'cafile' => $this->cafile
                    ]
                ]
            );
            $this->socket = stream_socket_client('tls://' .
$this->address . ':' . $this->port, $errno, $errstr, 60,
STREAM_CLIENT_CONNECT, $socketContext);
        } else {
            $this->socket = stream_socket_client('tcp://' .
$this->address . ':' . $this->port, $errno, $errstr, 60,
STREAM_CLIENT_CONNECT);
        }

        if (!$this->socket) {
            $this->_errorMessage("stream_socket_create()
$errno, $errstr");
            return false;
        }

        stream_set_timeout($this->socket, 5);
        stream_set_blocking($this->socket, 0);

        $i = 0;
        $buffer = '';

        $buffer .= chr(0x00);
        $i++; // Length MSB
        $buffer .= chr(0x04);
        $i++; // Length LSB
        $buffer .= chr(0x4d);
```

```
            $i++; // M
            $buffer .= chr(0x51);
            $i++; // Q
            $buffer .= chr(0x54);
            $i++; // T
            $buffer .= chr(0x54);
            $i++; // T
            $buffer .= chr(0x04);
            $i++; // // Protocol Level

            //No Will
            $var = 0;
            if ($clean) {
                $var += 2;
            }

            //Add will info to header
            if ($this->will !== null) {
                $var += 4; // Set will flag
                $var += ($this->will['qos'] << 3); //Set will qos
                if ($this->will['retain']) {
                    $var += 32;
                } //Set will retain
            }

            if ($this->username !== null) {
                $var += 128;
            }    //Add username to header
            if ($this->password !== null) {
                $var += 64;
            }    //Add password to header

            $buffer .= chr($var);
            $i++;

            //Keep alive
            $buffer .= chr($this->keepalive >> 8);
            $i++;
            $buffer .= chr($this->keepalive & 0xff);
            $i++;

            $buffer .= $this->strwritestring($this->clientid,
    $i);

            //Adding will to payload
            if ($this->will !== null) {
                $buffer .= $this->strwritestring($this-
    >will['topic'], $i);
                $buffer .= $this->strwritestring($this-
    >will['content'], $i);
            }

            if ($this->username !== null) {
                $buffer .= $this->strwritestring($this->username,
    $i);
            }
            if ($this->password !== null) {
```

```
                $buffer .= $this->strwritestring($this->password,
$i);
            }

            $head = chr(0x10);

            while ($i > 0) {
                $encodedByte = $i % 128;
                $i /= 128;
                $i = (int)$i;
                if ($i > 0) {
                    $encodedByte |= 128;
                }
                $head .= chr($encodedByte);
            }

            fwrite($this->socket, $head, 2);
            fwrite($this->socket, $buffer);

            $string = $this->read(4);

            if (ord($string[0]) >> 4 === 2 && $string[3] ===
chr(0)) {
                $this->_debugMessage('Connected to Broker');
            } else {
                $this->_errorMessage(
                    sprintf(
                        "Connection failed! (Error: 0x%02x
0x%02x)\n",
                        ord($string[0]),
                        ord($string[3])
                    )
                );
                return false;
            }

            $this->timesinceping = time();

            return true;
        }

        /**
         * Reads in so many bytes
         *
         * @param int $int
         * @param bool $nb
         *
         * @return false|string
         */
        public function read($int = 8192, $nb = false)
        {
            $string = '';
            $togo = $int;

            if ($nb) {
                return fread($this->socket, $togo);
            }
```

```
        while (!feof($this->socket) && $togo > 0) {
            $fread = fread($this->socket, $togo);
            $string .= $fread;
            $togo = $int - strlen($string);
        }

        return $string;
    }

    /**
     * Subscribes to a topic, wait for message and return it
     *
     * @param $topic
     * @param $qos
     *
     * @return string
     */
    public function subscribeAndWaitForMessage($topic, $qos):
string
    {
        $this->subscribe(
            [
                $topic => [
                    'qos' => $qos,
                    'function' => '__direct_return_message__'
                ]
            ]
        );

        do {
            $return = $this->proc();
        } while ($return === true);

        return $return;
    }

    /**
     * subscribes to topics
     *
     * @param $topics
     * @param int $qos
     */
    public function subscribe($topics, $qos = 0): void
    {
        $i = 0;
        $buffer = '';
        $id = $this->msgid;
        $buffer .= chr($id >> 8);
        $i++;
        $buffer .= chr($id % 256);
        $i++;

        foreach ($topics as $key => $topic) {
            $buffer .= $this->strwritestring($key, $i);
            $buffer .= chr($topic['qos']);
            $i++;
```

```
                $this->topics[$key] = $topic;
            }

            $cmd = 0x82;
            //$qos
            $cmd += ($qos << 1);

            $head = chr($cmd);
            $head .= $this->setmsglength($i);
            fwrite($this->socket, $head, strlen($head));

            $this->_fwrite($buffer);
            $string = $this->read(2);

            $bytes = ord(substr($string, 1, 1));
            $this->read($bytes);
        }

        /**
         * Sends a keep alive ping
         */
        public function ping(): void
        {
            $head = chr(0xc0);
            $head .= chr(0x00);
            fwrite($this->socket, $head, 2);
            $this->timesinceping = time();
            $this->_debugMessage('ping sent');
        }

        /**
         *  sends a proper disconnect cmd
         */
        public function disconnect(): void
        {
            $head = ' ';
            $head[0] = chr(0xe0);
            $head[1] = chr(0x00);
            fwrite($this->socket, $head, 2);
        }

        /**
         * Sends a proper disconnect, then closes the socket
         */
        public function close(): void
        {
            $this->disconnect();
            stream_socket_shutdown($this->socket,
    STREAM_SHUT_WR);
        }

        /**
         * Publishes $content on a $topic
         *
         * @param $topic
         * @param $content
         * @param int $qos
```

```
         * @param bool $retain
         */
        public function publish($topic, $content, $qos = 0,
    $retain = false): void
        {
            $i = 0;
            $buffer = '';

            $buffer .= $this->strwritestring($topic, $i);

            if ($qos) {
                $id = $this->msgid++;
                $buffer .= chr($id >> 8);
                $i++;
                $buffer .= chr($id % 256);
                $i++;
            }

            $buffer .= $content;
            $i += strlen($content);

            $head = ' ';
            $cmd = 0x30;
            if ($qos) {
                $cmd += $qos << 1;
            }
            if (empty($retain) === false) {
                ++$cmd;
            }

            $head[0] = chr($cmd);
            $head .= $this->setmsglength($i);

            fwrite($this->socket, $head, strlen($head));
            $this->_fwrite($buffer);
        }

        /**
         * Writes a string to the socket
         *
         * @param $buffer
         *
         * @return bool|int
         */
        protected function _fwrite($buffer)
        {
            $buffer_length = strlen($buffer);
            for ($written = 0; $written < $buffer_length;
    $written += $fwrite) {
                $fwrite = fwrite($this->socket, substr($buffer,
    $written));
                if ($fwrite === false) {
                    return false;
                }
            }
            return $buffer_length;
        }
```

```
    /**
     * Processes a received topic
     *
     * @param $msg
     *
     * @retrun bool|string
     */
    public function message($msg)
    {
        $tlen = (ord($msg[0]) << 8) + ord($msg[1]);
        $topic = substr($msg, 2, $tlen);
        $msg = substr($msg, ($tlen + 2));
        $found = false;
        foreach ($this->topics as $key => $top) {
            if (preg_match(
                '/^' . str_replace(
                    '#',
                    '.*',
                    str_replace(
                        '+',
                        "[^\/]*",
                        str_replace(
                            '/',
                            "\/",
                            str_replace(
                                '$',
                                '\$',
                                $key
                            )
                        )
                    )
                ) . '$/',
                $topic
            )) {
                $found = true;

                if ($top['function'] ===
'__direct_return_message__') {
                    return $msg;
                }

                if (is_callable($top['function'])) {
                    call_user_func($top['function'], $topic,
$msg);
                } else {
                    $this->_errorMessage('Message received on
topic ' . $topic . ' but function is not callable.');
                }
            }
        }

        if ($found === false) {
            $this->_debugMessage('msg received but no match
in subscriptions');
        }
```

```
        return $found;
    }

    /**
     * The processing loop for an "always on" client
     * set true when you are doing other stuff in the loop
good for
     * watching something else at the same time
     *
     * @param bool $loop
     *
     * @return bool | string
     */
    public function proc(bool $loop = true)
    {
        if (feof($this->socket)) {
            $this->_debugMessage('eof receive going to
reconnect for good measure');
            fclose($this->socket);
            $this->connect_auto(false);
            if (count($this->topics)) {
                $this->subscribe($this->topics);
            }
        }

        $byte = $this->read(1, true);

        if ((string)$byte === '') {
            if ($loop === true) {
                usleep(100000);
            }
        } else {
            $cmd = (int)(ord($byte) / 16);
            $this->_debugMessage(
                sprintf(
                    'Received CMD: %d (%s)',
                    $cmd,
                    isset(static::$known_commands[$cmd]) ===
true ? static::$known_commands[$cmd] : 'Unknown'
                )
            );

            $multiplier = 1;
            $value = 0;
            do {
                $digit = ord($this->read(1));
                $value += ($digit & 127) * $multiplier;
                $multiplier *= 128;
            } while (($digit & 128) !== 0);

            $this->_debugMessage('Fetching: ' . $value . '
bytes');

            $string = $value > 0 ? $this->read($value) : '';

            if ($cmd) {
                switch ($cmd) {
```

```
                        case 3: //Publish MSG
                            $return = $this->message($string);
                            if (is_bool($return) === false) {
                                return $return;
                            }
                            break;
                    }
                }
            }

            if ($this->timesinceping < (time() - $this-
>keepalive)) {
                $this->_debugMessage('not had something in a
while so ping');
                $this->ping();
            }

            if ($this->timesinceping < (time() - ($this-
>keepalive * 2))) {
                $this->_debugMessage('not seen a packet in a
while, disconnecting/reconnecting');
                fclose($this->socket);
                $this->connect_auto(false);
                if (count($this->topics)) {
                    $this->subscribe($this->topics);
                }
            }
            return true;
        }

        /**
         * Gets the length of a msg, (and increments $i)
         *
         * @param $msg
         * @param $i
         *
         * @return float|int
         */
        protected function getmsglength(&$msg, &$i)
        {
            $multiplier = 1;
            $value = 0;
            do {
                $digit = ord($msg[$i]);
                $value += ($digit & 127) * $multiplier;
                $multiplier *= 128;
                $i++;
            } while (($digit & 128) !== 0);

            return $value;
        }

        /**
         * @param $len
         *
         * @return string
         */
```

```
    protected function setmsglength($len): string
    {
        $string = '';
        do {
            $digit = $len % 128;
            $len >>= 7;
            // if there are more digits to encode, set the
top bit of this digit
            if ($len > 0) {
                $digit |= 0x80;
            }
            $string .= chr($digit);
        } while ($len > 0);
        return $string;
    }

    /**
     * @param $str
     * @param $i
     *
     * @return string
     */
    protected function strwritestring($str, &$i): string
    {
        $len = strlen($str);
        $msb = $len >> 8;
        $lsb = $len % 256;
        $ret = chr($msb);
        $ret .= chr($lsb);
        $ret .= $str;
        $i += ($len + 2);
        return $ret;
    }

    /**
     * Prints a sting out character by character
     *
     * @param $string
     */
    public function printstr($string): void
    {
        $strlen = strlen($string);
        for ($j = 0; $j < $strlen; $j++) {
            $num = ord($string[$j]);
            if ($num > 31) {
                $chr = $string[$j];
            } else {
                $chr = ' ';
            }
            printf("%4d: %08b : 0x%02x : %s \n", $j, $num,
$num, $chr);
        }
    }

    /**
     * @param string $message
     */
```

```
        protected function _debugMessage(string $message): void
        {
            if ($this->debug === true) {
                echo date('r: ') . $message . PHP_EOL;
            }
        }

        /**
         * @param string $message
         */
        protected function _errorMessage(string $message): void
        {
            error_log('Error:' . $message);
        }
}
```

f. Transaksi. Php

```
<?php

include __DIR__."/../config/database.php";

class Transaksi{

function parkir(){

global $conn;

return $conn->query("SELECT * FROM transaksi WHERE
status='IN'");

}

function keluar(){

global $conn;

return $conn->query("SELECT * FROM transaksi WHERE
status='OUT'");

}

function riwayat(){

global $conn;

return $conn->query("SELECT * FROM transaksi WHERE
status='DONE'");

}

}

?>
```

g. Uji coba

Pengujian login

No	Scenario pengujian	Test case	Hasil yang di harapkan	Hasil pengujian	kesimpulan
1	Username dan password kosong	Klik tombol login	Sistem menolak dan menampilkan pesan "Harap isi username dan password"	Sesuai harapan	valid
2	Username benar, password salah	Login dengan password salah	Sistem menampilkan pesan "password atau username salah"	Sesuai harapan	valid
3	Username dan password benar	Klik tombol login	Sistem berhasil masuk ke dashboard	Sesuai harapan	valid

Pengujian Check-in parkir

No	Scenario pengujian	Test case	Hasil yang di harapkan	Hasil pengujian	kesimpulan
1	Tap kartu RFID	Pengguna menempelkan kartu pada reader	System membaca kartu dan mencatat kendaraan masuk	Sesuai harapan	Valid

Pengujian Check-out parkir

No	Scenario pengujian	Test case	Hasil yang di harapkan	Hasil pengujian	kesimpulan
1	Tap kartu keluar	Pengguna menempelkan kartu Kembali pada reader	System mencatat jam keluar dan menghitung biaya parkir	Sesuai harapan	Valid

Pengujian Logout

No	Scenario pengujian	Test case	Hasil yang di harapkan	Hasil pengujian	kesimpulan
1	Logout aplikasi	Klik tombol logout	Sistem keluar dari aplikasi dan Kembali ke halaman login	Sesuai harapan	Valid

h. Link GitHub

https://github.com/cello1833/UKK-SMART-PARKING-